Practice Test 12 — Answer Key

Comprehensive Final Exam Preparation (Modules 01–17)

Part A: Multiple Choice

1. **A** — All living things have cellular organization — this is universal to life
2. **B** — Cell → Tissue → Organ → Organ System → Organism is the correct hierarchy
3. **C** — The independent variable is what the scientist deliberately changes
4. **D** — A theory is well-supported by extensive, repeated experimental evidence
5. **A** — Homeostasis = maintaining internal stability despite external changes
6. **B** — Atomic number = number of protons in the nucleus
7. **C** — Covalent bonds involve sharing electrons between atoms
8. **D** — Oxygen pulls shared electrons more strongly → unequal sharing → polarity
9. **A** — pH 3 is well below 7, making it strongly acidic
10. **B** — Carbon, Hydrogen, Oxygen, Nitrogen make up ~96% of living matter
11. **C** — Dehydration synthesis removes water to join monomers into polymers
12. **D** — Carbohydrates (glucose) are the body's preferred quick energy source
13. **A** — Amino acids are the monomers of proteins (20 different types)
14. **B** — The amino acid sequence dictates how the protein folds and what it does
15. **C** — DNA and RNA are nucleic acids built from nucleotide monomers

16. **D** — Cell theory: all living things are made of cells; all cells come from pre-existing cells
17. **A** — Prokaryotes lack a membrane-bound nucleus (pro = before, karyon = nucleus)
18. **B** — Mitochondria are the "powerhouse of the cell" — site of aerobic respiration
19. **C** — Ribosomes translate mRNA into proteins (protein synthesis)
20. **D** — Plant cells have a rigid cell wall made of cellulose; animal cells do not
21. **A** — Integral membrane proteins move laterally in the lipid bilayer—the "fluid mosaic."
22. **B** — Hydrophobic tails cluster inward, away from aqueous environments.
23. **C** — Cholesterol buffers membrane fluidity across temperature changes.
24. **D** — Integral (transmembrane) proteins span from one surface to the other.
25. **A** — Hypotonic surroundings cause net water influx—red blood cells swell and may lyse.
26. **B** — Metabolism includes all chemical reactions (anabolism + catabolism).
27. **C** — Exergonic reactions release energy when bonds are rearranged.
28. **D** — Enzymes speed reactions by lowering activation energy.
29. **A** — The substrate binds at the enzyme active site.
30. **B** — Energy cannot be created or destroyed—only transformed/moved.
31. **C** — DNA (deoxyribonucleic acid) carries the genetic instructions.
32. **D** — The four DNA bases are Adenine, Thymine, Guanine, and Cytosine.
33. **A** — In DNA, Adenine pairs with Thymine (A–T).
34. **B** — The backbone is made of alternating sugar (deoxyribose) and phosphate groups.

35. **C** — Each new DNA molecule keeps one original strand and adds one newly built strand.

36. **D** — DNA is copied during S (Synthesis) phase of interphase.

37. **A** — Prophase → Metaphase → Anaphase → Telophase (remember: PMAT).

38. **B** — Chromosomes line up at the equator of the cell during Metaphase.

39. **C** — Mitosis produces two genetically identical diploid cells.

40. **D** — Mitosis is for growth, repair, and replacement of body (somatic) cells.

41. **A** — Homeostasis is the maintenance of a stable internal environment.

42. **B** — A thermostat is a classic example of negative feedback: the response (heating) opposes the change (cold) to restore the set point.

43. **C** — Osmoregulation is specifically the regulation of water and salt balance.

44. **D** — The urinary (excretory) system, primarily the kidneys, regulates water and salt levels.

45. **A** — Chewing is mechanical digestion — physically breaking food into smaller pieces without changing its chemistry.

46. **B** — An allele is one of two or more alternate forms of a gene (e.g., T for tall, t for short).

47. **C** — Genotype is the genetic makeup (e.g., Tt); phenotype is the observable trait (e.g., tall).

48. **D** — Two identical alleles (BB or bb) = homozygous.

49. **A** — Two different alleles (Bb) = heterozygous.

50. **B** — Gregor Mendel studied inheritance using pea plants (Pisum sativum).

51. **C** — The skeletal system supports, protects, enables movement, stores minerals, and produces blood cells — but it does not secrete digestive hormones.
52. **D** — Red bone marrow is the site of hematopoiesis (production of red blood cells, white blood cells, and platelets).
53. **A** — Osteoclasts break down bone matrix, releasing stored calcium into the bloodstream.
54. **B** — The diaphysis is the shaft (long central portion) of a long bone.
55. **C** — The epiphyses are the rounded ends of a long bone, where joints form.
56. **D** — Skeletal muscles are attached to bones and produce voluntary movement; contraction also generates heat.
57. **A** — When muscles contract, ATP is used and heat is released as a byproduct, helping maintain body temperature.
58. **B** — Whole muscle → fascicle (bundle of fibers) → muscle fiber (single cell) → myofibril (contractile strand within the fiber).
59. **C** — The sarcomere is the repeating unit between two Z-lines; it is the basic contractile unit of striated muscle.
60. **D** — Thick filaments are composed of the motor protein myosin.
61. **A** — The CNS is the brain and spinal cord — the integration and command center.
62. **B** — The PNS consists of all nerves and ganglia outside the brain and spinal cord, connecting the CNS to the rest of the body.
63. **C** — The somatic nervous system controls voluntary (conscious) movements of skeletal muscles.
64. **D** — The sympathetic division activates the "fight-or-flight" response, increasing heart rate, dilating pupils, and redirecting blood to muscles.

65. **A** — Dendrites are the branched extensions that receive signals from other neurons or sensory receptors.

66. **B** — Prokaryotic cells lack a membrane-bound nucleus; their DNA is in the nucleoid region.

67. **C** — Most bacteria have a single circular chromosome located in the nucleoid (not enclosed by a membrane).

68. **D** — Cocci are spheres, bacilli are rods, and spirilla are spirals — the three basic bacterial morphologies.

69. **A** — Gram-positive bacteria have a thick peptidoglycan wall exposed to the outside and no outer membrane.

70. **B** — Gram-positive bacteria retain the crystal violet dye during the Gram stain procedure and appear purple/violet.

71. **C** — Deoxygenated blood from the systemic veins drains into the **right atrium** first (superior and inferior vena cava), before passing to the right ventricle.

72. **D** — The right ventricle ejects blood into the **pulmonary trunk**, which branches into the **pulmonary arteries** carrying blood to the lungs.

73. **A** — The first heart sound ("lub," S1) is mainly the closing of the **AV (atrioventricular) valves** between atria and ventricles at the start of ventricular systole.

74. **B** — The second heart sound ("dup," S2) is mainly the closing of the **semilunar valves** (pulmonary and aortic) at the end of ventricular systole.

75. **C** — **Arteries** carry blood **away** from the heart (whether oxygen-rich or oxygen-poor depends on which artery).

76. **D** — Producers capture light energy and build organic molecules that feed other trophic levels.

77. **A** — Energy and biomass typically pass producer → primary consumer → higher consumers.

78. **B** — Biotic factors are living; abiotic factors are non-living chemical/physical conditions.

79. **C** — Materials recycle through biogeochemical cycles; energy dissipates as heat.

80. **D** — Carrying capacity (K) reflects limiting resources and space.

81. **A** — Selection requires variation, inheritance, and differences in survival/reproduction.

82. **B** — Homologous traits descend from a common ancestor; analogous traits converge by similar roles.

83. **C** — Chance allele-frequency shifts matter most when sampling error is large (small N).

84. **D** — Barriers to gene flow allow divergence—prezygotic or postzygotic mechanisms.

85. **A** — Transitional similarities in anatomy align with descent with modification.

Part B: Fill in the Blank

86. Accept **`observation`** (also `Observation`, `experimental observation`).

87. Accept **`organism`** (also `organism`).

88. Accept **`covalent`** (also `covalent`).

89. Accept **`acidic`** (also `acidic`, `acid`).

90. Accept **`protein`** (also `protein`).

91. Accept **`nitrogenous`** (also `nitrogenous`).

92. Accept **`mitochondria`** (also `mitochondria`, `mitochondrion`).

93. Accept **`nucleus`** (also `nucleus`).

94. Accept **`down`** (also `down`).

95. Accept **`osmosis`** (also `osmosis`).

96. Accept **`respiration`** (also `respiration`, `cellular respiration`).

97. Accept **`glucose`** (also `glucose`).

98. Accept **`template`** (also `template`).

99. Accept **`mRNA`** (also `mRNA`, `messenger RNA`).

100. Accept **`anaphase`** (also `anaphase`).

101. Accept **`meiosis`** (also `meiosis`).

102. Accept **`barriers`** (also `barriers`, `barrier`).

103. Accept **`collagen`** (also `collagen`).

104. Accept **`heterozygous`** (also `heterozygous`).

105. Accept **`sex`** (also `sex`, `X and Y`, `X/Y`).

106. Accept **`osteoclasts`** (also `osteoclasts`, `osteoclast`).

107. Accept **`synovial`** (also `synovial`).

108. Accept **`myosin`** (also `myosin`).

109. Accept **`fibers`** (also `fibers`, `fibres`).

110. Accept **`potentials`** (also `potentials`, `potential`).

111. Accept **`cleft`** (also `cleft`).

112. Accept **`membrane`** (also `membrane`).

113. Accept **`plasmids`** (also `plasmids`, `plasmid`).

114. Accept **`aorta`** (also `aorta`).

115. Accept `capillaries` (also `capillaries`, `capillary`).

116. Accept `trophic` (also `trophic`).

117. Accept `resources` (also `resources`, `resource`).

118. Accept `variation` (also `variation`).

119. Accept `isolation` (also `isolation`, `reproductive isolation`).

Part C: Short Answer / Free Response

120. Sample points for full credit: - Observation records measurable evidence; opinion states belief without required evidence.

121. Sample points for full credit: - Stable temperature/pH/blood chemistry keeps enzymes near optimal activity ranges.

122. Sample points for full credit: - Ionic: electron transfer forming ions attracted together (NaCl). Covalent: shared pairs (water O–H).

123. Sample points for full credit: - Lower pH means higher hydrogen ion concentration—more acidic.

124. Sample points for full credit: - Specific 3-D active site fits substrates; denatured enzyme loses catalytic shape.

125. Sample points for full credit: - Linear/base-paired sequences encode digital-like instructions sugars alone lack.

126. Sample points for full credit: - Mitochondria oxidize fuel aerobically; chloroplasts trap light energy in plants/algae.

127. Sample points for full credit: - Both translate mRNA with ribosomal machinery—universal genetic code.

128. Sample points for full credit: - Passive: diffusion/facilitated down gradients; Active: pumps move against gradients (Na^{+} /K^{+} pump).

129. Sample points for full credit: - Hypertonic medium draws water out—cells shrink (crenation for RBCs).

130. Sample points for full credit: - Maintenance transport, biosynthesis, signaling, repair—even Basal metabolic demand consumes ATP.

131. Sample points for full credit: - Aerobic uses oxygen as final electron acceptor; fermentation regenerates NAD^{+} without O_2.

132. Sample points for full credit: - DNA replication/transcription → RNA processing/export → translation → folded protein.

133. Sample points for full credit: - Mutation is DNA sequence change—provides variation raw material for evolution.

134. Sample points for full credit: - Mitosis clones somatic genome; Meiosis halves homologues for gametes.

135. Sample points for full credit: - Chiasmata exchange segments—new allele combos on chromosomes.

136. Sample points for full credit: - Examples: epithelium lining airways; bone/cartilage connective support; skeletal muscle moves bones.

137. Sample points for full credit: - Cartilage flexible matrix resists compression at joints; rigid bone handles torque.

138. Sample points for full credit: - Dominant healthy allele masks recessive disease allele in heterozygotes.

139. Sample points for full credit: - Pedigrees show inheritance patterns across families; Punnett predicts offspring ratios.

140. Sample points for full credit: - Osteoclasts remove damaged bone; osteoblasts lay new matrix—Wolff's law adaptation.

141. Sample points for full credit: - Synovial fluid secreted into cavity lubricates cartilage surfaces.

142. Sample points for full credit: - Ca^{2+} binds troponin → tropomyosin shift exposes actin sites for myosin.

143. Sample points for full credit: - Additional fibers recruited until fused tetanus needed—size principle partially.

144. Sample points for full credit: - Resting negative inside becomes less negative during Na^{+} influx—depolarization phase.

145. Sample points for full credit: - Sympathetic raises HR/stress; parasympathetic lowers HR/digest emphasis.

146. Sample points for full credit: - Full dosing kills susceptible bacteria—less chance survivors mutate resistance.

147. Sample points for full credit: - Bacteria metabolize independently divide; viruses need host machinery.

148. Sample points for full credit: - Pulmonary: right heart→lungs for oxygen pick-up; Systemic: left heart→body delivery.

149. Sample points for full credit: - High alveolar PO_2 loads hemoglobin; elevated tissue PCO_2 favors CO_2 diffusion out.

150. Sample points for full credit: - Energy enters as sunlight→ producers → herbivores → carnivores; nutrients cycle biogeochemically.

151. Sample points for full credit: - K reflects limits—factors like drought/predation reduce carrying capacity.

152. Sample points for full credit: - Traits inherited vary; some reproduce more—allele frequencies shift generations.

153. Sample points for full credit: - Examples: temporal mating seasons, geographic barriers, gametic incompatibility reducing gene flow.